

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1518

COURSE OUTCOME COVERED

CO3: Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Code of Conduct (19247223)

I G. Jaswanth (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

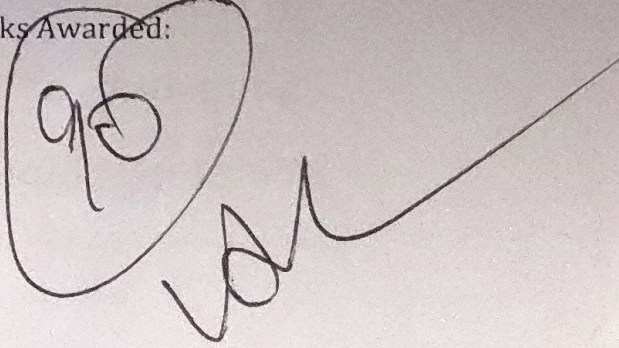
Signature of the student: G. Jaswanth

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

95



Assessment Tasks

Experiment 1: Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg no of the student , Subjects , Marks, Average, percentage, rank, overall performance etc.

Experiment 2: Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Buy, Rent, commercial, Rental Agreement , Rental Agreement, Area, place, Range, property value etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

Exp 1: Mark Sheet Processing system using SaaS.

Aim:

To create and deploy a simple cloud-based Mark sheet Processing system using a cloud service provider to demonstrate SaaS.

Algorithm:

- ① Create a simple web application using HTML, CSS and Javascript
- ② Create a Project in a cloud service as Firebase
- ③ Configure cloud Firestore as the cloud database
- ④ Create input fields for:
 - Student Name
 - Register Number
 - Subject Names
 - Marks
- ⑤ Implement Javascript functions to calculate: total, average, percentage and rank.
- ⑥ Display the overall performance
- ⑦ Store the result in the cloud.

Implementation:

Mark sheet Processing	User details	
Users Details ✓	Name	Jashwanth Gopinreddy
Users details	Phone	+91 6309765009
All User details	Email	192472235_smarth@saareetha.com
	Date-Time	20-08-2026
	Theatres	
	Drop Down	- select -
		Submit Reset

Output :

Name : G. Jaswanth
Register No : 192472235
Total : 358
Average : 89.5
Percentage : 89.5%
Performance : Excellent.

Enf 2 - Property Buying & Rental system

Aim :

To develop a simple cloud-based property buying and rental system using SaaS

Procedure :

1. Open the cloud-based property application
2. Select Buy or Rent
3. Select Property type
4. Enter location, area, and price range
5. Display the matching property details
6. Store the property information in the cloud.

Implementation :

Property Management	Property Details
Properties	Transaction Type Rent ▼
Search	Property Type Residential ▼
Buy	Location OMR, Chennai
Rent	Area 1200 sq.ft
Agreement	Property value ₹ 60,00,000
	Monthly Rent ₹ 25,000
	Price Range ₹ 20,000 - ₹ 30,000
	Rental Agreement 11 Months

put :

Property Details

Transaction : RENT
Property Type : Residential
Location : OMR, Chennai
Area : 1200 sq.ft
Property value : ₹ 60,00,000
Monthly Rent : ₹ 25,000
Agreement : 11 Months

Exp 3 - Type 2 virtualization

Aim : To create and configure a virtual machine using a Type-2 Hypervisor with specified CPU, RAM and storage.

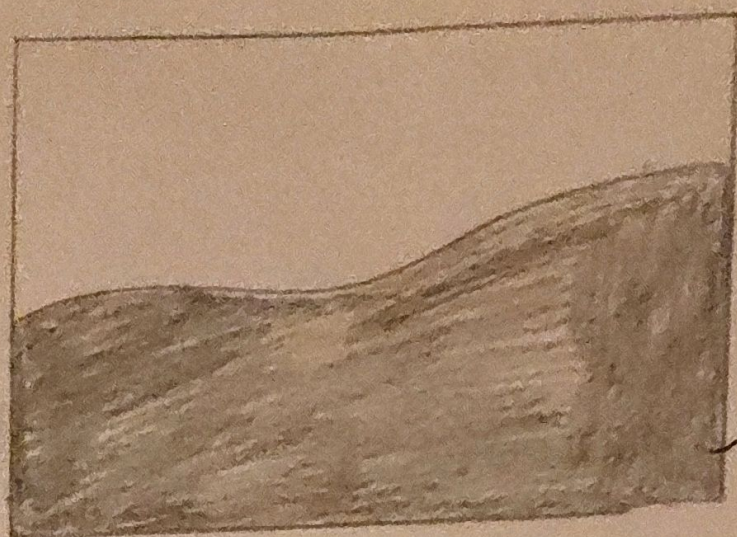
Procedure :

- ① open Oracle Virtual Box / VMware
- ② create a new virtual machine
- ③ select the guest OS
- ④ configure CPU, RAM and storage
- ⑤ Attach the OS ISO file
- ⑥ start the virtual machine
- ⑦ verify the VM configuration.

Implementation :

Parameter	Value
Hypervisor	VMware Workstation
VM Name	Cloud-VM
Guest OS	Ubuntu Linux
CPU	5 Cores
RAM	12GB
Storage	8GB

Output :



Exp 4 : VM snapshot and cloning

Aim :

To create and restore a snapshot and clone of a virtual machine using a Type-2 Hypervisor.

Procedure :

- ① Open the existing VM in VirtualBox
- ② Start the VM and create an initial state
- ③ Take a snapshot
- ④ Make changes to the VM.
- ⑤ Restore the previous snapshot
- ⑥ Verify that the previous state is recovered
- ⑦ Clone the VM
- ⑧ Start and verify the cloned VM.

Implementation :

Parameter	Value
VM Name	Cloud-VM
Snapshot Name	Initial_State
Test Folder	Experiment_1
Modified Folder	Experiment_2
Clone Name	Cloud-VM_Cloned

Before snapshot:

Create → Experiment_1

Take snapshot:

Initial_state

After snapshot:

Create → Experiment_2

Restore:

Initial_state

Output:

Snapshot : Initial_state

Snapshot : Created successfully

Restore : Successful

Previous : Recovered
state

clone : cloud-vm-clone

clone status: Running successfully.

